Aprendizagem Automática (APRAU)

Mestrado em Engenharia Informática

Assignment

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1 Objectives

The **general objective** of this assignment is to apply different machine learning methods on a multi-disease clinical dataset, analyze model behavior, and critically interpret the results obtained.

2 Exploratory Data Analysis (EDA)

The dataset you will be working with is derived from a consolidated medical data warehouse designed to study comorbidity patterns across chronic cardiometabolic diseases. It includes demographic profiles, anthropometric measurements, lifestyle indicators, and clinical laboratory biomarkers collected from healthcare screening programs.

Each instance in the dataset represents a single patient profile with two associated target variables: a categorical label indicating the clinical diagnostic category (`target_class`, used for classification tasks) and a continuous laboratory measurement (`target_regression`, used for regression tasks).

Features

The dataset contains a rich set of demographic, behavioral, and clinical predictors:

- **age**: Patient chronological age in years;
- **age_normalized**: Standardized min-max scaling of the patient's age;
- **age_level**: Derived age category (*Young*, *Middle*, *Mature*, *Senior*);
- **gender**: Biological sex (*Male*, *Female*);
- **bmi**: Body Mass Index in kg/m^2;
- **bmi_level**: WHO-standardized nutritional status (*Underweight*, *Normal*, *Overweight*, *Obese*);
- **smoking**: Standardized tobacco consumption status (*Never*, *Former*, *Current*);
- **physical_activity**: Self-reported weekly exercise intensity (*Low*, *Moderate*, *High*);
- **stress_level**: Quantified psychological stress tier (*Low*, *Medium*, *High*);
- **sleep_hours**: Average self-reported daily sleep duration;
- **alcohol_intake**: Quantified metric of habitual alcohol consumption;
- **salt_intake**: Dietary sodium and salt intake estimate;
- **sugar_consumption**: Frequency of dietary refined sugar intake (*Low*, *Medium*, *High*);
- **family_history**: Known familial predisposition to chronic cardiometabolic conditions (*Yes*, *No*);
- **systolic_bp**: Systolic blood pressure reading in mmHg;
- **diastolic_bp**: Diastolic blood pressure reading in mmHg;

- **blood_pressure**: Aggregated arterial tension measurement;
- **heart_rate**: Resting cardiac frequency in beats per minute (bpm);
- **HbA1c_level**: Glycated hemoglobin percentage (marker of medium-term glycemic control);
- **cholesterol**: Total serum cholesterol concentration;
- **triglycerides**: Serum triglyceride level;
- **hdl**: High-Density Lipoprotein ("good" cholesterol level);
- **ldl**: Low-Density Lipoprotein ("bad" cholesterol level);
- **low_hdl_cholesterol**: Binary clinical indicator flag for sub-optimal HDL levels;
- **high_ldl_cholesterol**: Binary clinical indicator flag for elevated LDL risk;
- **crp_level**: Serum C-Reactive Protein concentration (systemic vascular inflammation marker);
- **homocysteine_level**: Plasma homocysteine concentration (vascular and metabolic risk indicator);
- **education_level**: Highest educational attainment (*Primary*, *Secondary*, *Tertiary*);
- **employment_status**: Current professional status (*Employed*, *Unemployed*, *Retired*);
- **systolic_index**: Secondary index capturing hemodynamic pressure dynamics;
- **calcium_baseline**: Biochemical electrolyte baseline recording;
- **plasma_value**: Measured plasma viscosity variance indicator;
- **ldl_alternative**: Alternative lipid particle profile representation;
- **target_class**: Categorical diagnosis sublabel representing the patient's condition;
- **target_regression**: Continuous laboratory biomarker concentration.

Dataset Assignment and Initial Tasks

Each group will be provided with a curated subset of **2,000 patient records**, rather than the full data dataset.

Each dataset containing **three different diagnostic classes** (`target_class`), with each group being assigned a different combination of classes. As the number of samples varies across classes, most groups will work with an **imbalanced dataset**.

After receiving the dataset, perform an exploratory data analysis to obtain more information from the data:

- **Descriptive Statistics**;
- **Univariate Analysis** (Distribution of individual features);
- **Bivariate Analysis** (Correlation between features and the different target variables).

Perform univariate and bivariate analyses to identify relevant data characteristics and relationships between features. Identify potentially uninformative, constant, or redundant features, and justify your findings based on the results obtained.

3 Methods Application - Regression

Using the provided dataset, build two regression models to predict `target_regression`. You may use the holdout method to evaluate the models.

- **Simple Linear Regression**: Fit a model using a single feature. Test different features, evaluate their performance, and select the most suitable one.
- **Multiple Linear Regression**: Fit a model using several clinical features. Experiment with different combinations of features and determine the best-performing group of features.
- **Comparison and Discussion**: Evaluate both models using appropriate metrics (R^2, MAE, RMSE). Compare their performance and provide a short discussion of the results.

4 Methods Application - Classification

Consider using the following methods: **Logistic Regression** and **Discriminant Analysis (LDA)**. Applying the methods to the chosen data, decide which method is most appropriate for the problem, giving reasons for your choice. Use the following resampling methods:

- Holdout;
- Cross Validation (with $k = 5$ and $k = 10$);
- Leave-One-Out Cross Validation (LOOCV);
- Bootstrap.

Use the evaluation metrics that you find most appropriate to evaluate the results obtained in each experiment. Analyzing the results obtained, indicate how the variance is affected by the resampling methods used.

5 Feature Selection

Evaluate the impact of feature selection on classification performance using regularization methods (Ridge and Lasso). Compare the results obtained using all available features with those obtained using a reduced set of features, and critically discuss the differences.

6 Submissions

A notebook with answers to the proposed tasks. The notebook is .ipynb by default. Any other format must be easily readable. Please take care with the following:

- The steps taken must be succinctly and adequately described (through comments in the code or text cells in the notebook)
- The results must be rigorously summarized and clearly presented as much as possible.

6.1 Groups

- Assignments are submitted by groups of 3 students (exceptionally, groups of 2 elements may be accepted). Different elements may have different grades based on the contribution distribution and interactions about the assignment.
- Code of Conduct
  - All the materials used and consulted must be credited in the work as references.

  - All students should know the Disciplinary Regulations for Students of Polytechnic Institute of Porto (https://dre.pt/dre/detalhe/despacho/4103-2013-2301392)

- It is mandatory the Github version control tool. Each group must share the repository with PL teacher.

6.2 Submissions and Deadlines

There are **two mandatory submissions** via Moodle:

- **2 November**: Intermediate submission (25% of the final grade).
- **20 December**: Final submission (45% of the final grade).

Only submissions made through Moodle before the specified deadlines will be accepted. Late submissions will not be considered.

6.3 Submission Format

The ZIP file must be named `APRAU_AAA_CCC_Num1_Num2_Num3.zip`, where `AAA` is the teacher's acronym, `CCC` is the class identifier, and `Num1`, `Num2`, and `Num3` are the student numbers of the group members.

6.4 Presentation and Assessment

- The presentation and defense are **mandatory for all group members** and will take place on a date scheduled by the PL teacher (see the APRAU course syllabus).
- **The project grade is individual**, with the individual defense accounting for **70% of the project grade**, placing particular emphasis on the knowledge and skills effectively demonstrated by each student.